HCM-UTE

Edge Computing Innovation

TRINITY

REAL-TIME CONSTRUCTION SITE SAFETY MONITORING SYSTEM BASED ON EDGE COMPUTING AND COMPUTER VISION

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PROJECT OUTLINE

1. Motivation
2. Objective
3. Related work & Contribution
4. Dataset
5. Method
6. Results
7. Conclusion
8. Application demo

1. Motivation

- Construction sites are highly dangerous environments that require strict safety monitoring.
- 7,004 occupational accidents / 658 fatalities in Vietnam, 2025 (MOLISA)
- Manual CCTV monitoring suffers fatigue, misses real-time violations
- Cloud AI: high latency, heavy bandwidth cost, privacy risk

PPE violated construction site

2. Objective

- Develop an Edge AI system for real-time construction site safety monitoring.
- Detect PPE violations and hazardous situations automatically.
- Provide low-latency on-site alerts.
- Reduce false alarms through temporal verification.
- Minimize bandwidth and preserve privacy using an edge-first architecture.

3. Related Work & Contributions

Related Works	Research Gaps
YOLO-based PPE detection with attention modules **[Z. Wu et al, 2025]**	Object-focused detection without hazard context.
Edge computing for construction monitoring **[X. Xiao et al., 2024 & M. Gugssa et al., 2024]**	Bandwidth savings mainly reported as theoretical estimates.
Frame-by-frame vision monitoring **[S. JoonOh et al.]**	High false-positive rate caused by transient detections.
Model compression for edge AI **[B. Tong et al.]**	Limited validation in real-world deployment scenarios.

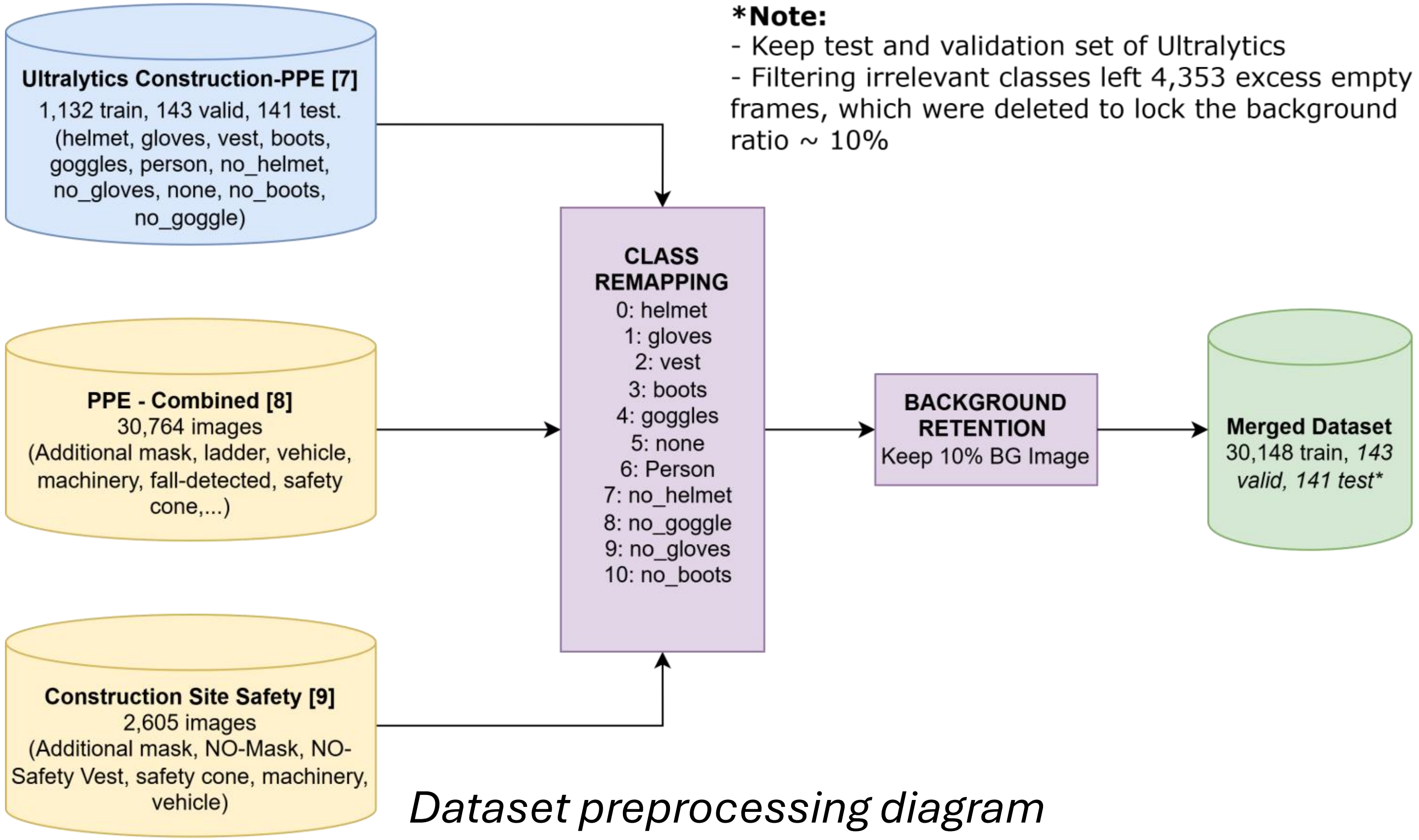
3. Related Work & Contributions

- **Context-Aware Alerting:** Worker location with PPE status to generate zone-aware WARNING/CRITICAL alerts.
- **Bandwidth Efficiency:** Reduced upstream traffic by **99.93% (1.35 MB/hour).**
- **Temporal Smoothing:** Five-frame confirmation reduced false alarms by **94.6%**, maintaining **167 ms** alert latency
- **Deployment Benchmark:** Validated the ONNX FP32 deployment pipeline, achieving **131 FPS** in an edge-oriented GPU proxy environment

Contributions

4. Dataset

The **Ultralytics Construction-PPE** dataset was used as the base dataset. Two additional public datasets were merged through class remapping to increase training samples and improve class balance.



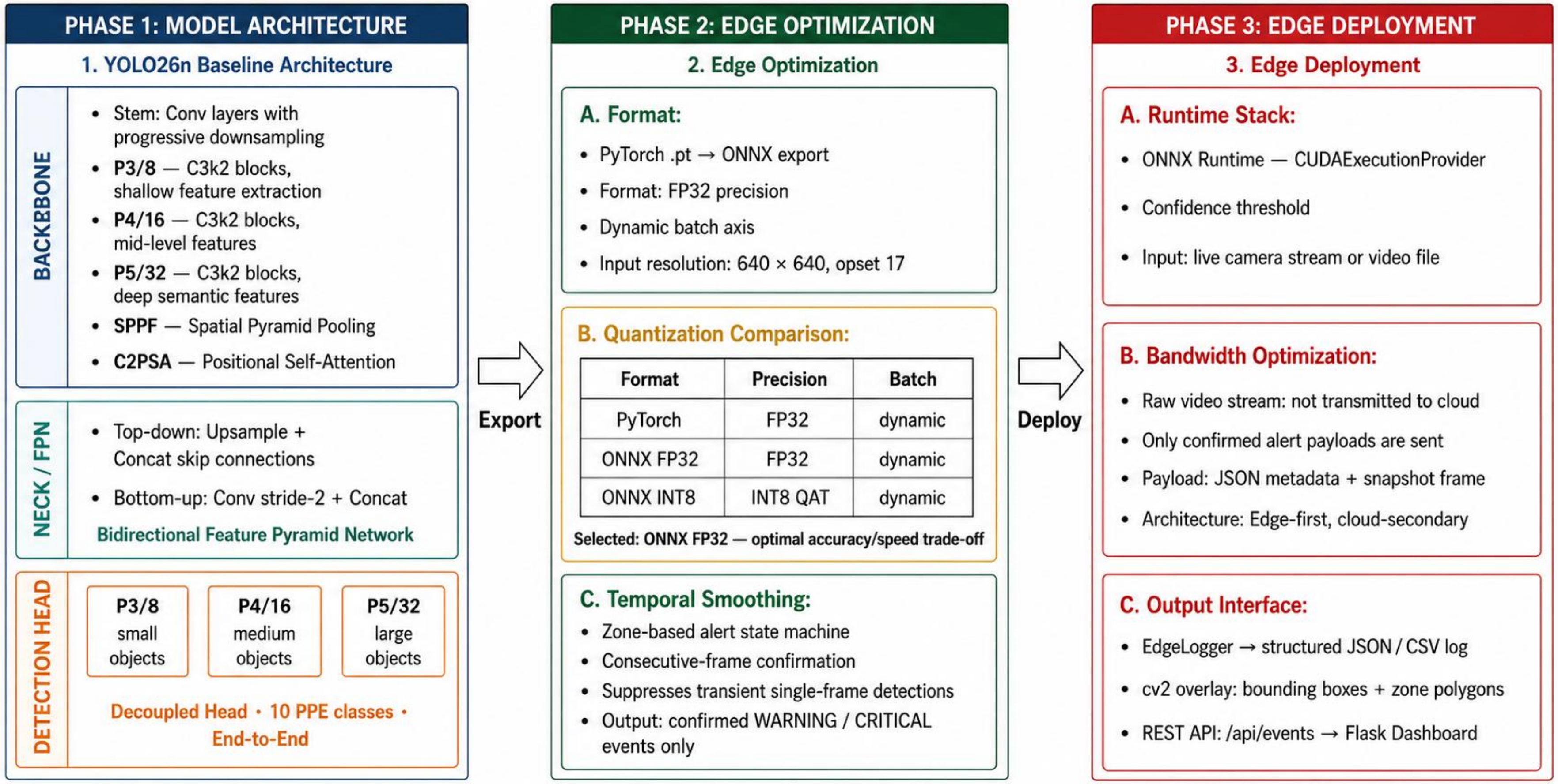
Dataset preprocessing diagram

Class Name	Ultralytics Train Images	Merged Train Instances
helmet	1357	33,482
gloves	1162	4,527
vest	1283	8,801
boots	1251	1,235
goggles	427	3,378
none	654	651
Person	1790	12,336
no_helmet	400	12,423
no_goggle	337	3,154
no_gloves	442	4,722
no_boots	88	88

Original and merged dataset class distribution

5. Method

5.1. Overall method



Overall method of the project

5.2. Workflow

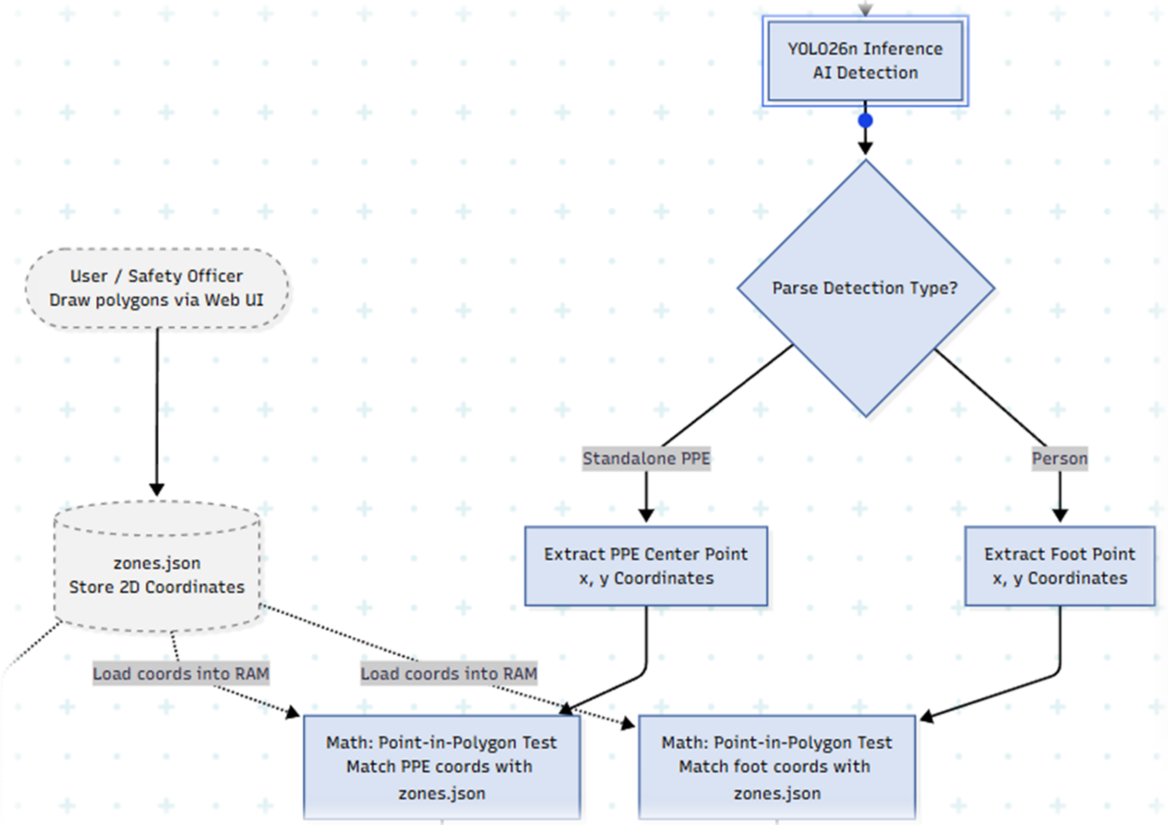
PHASE 1: INPUT & AI DETECTION

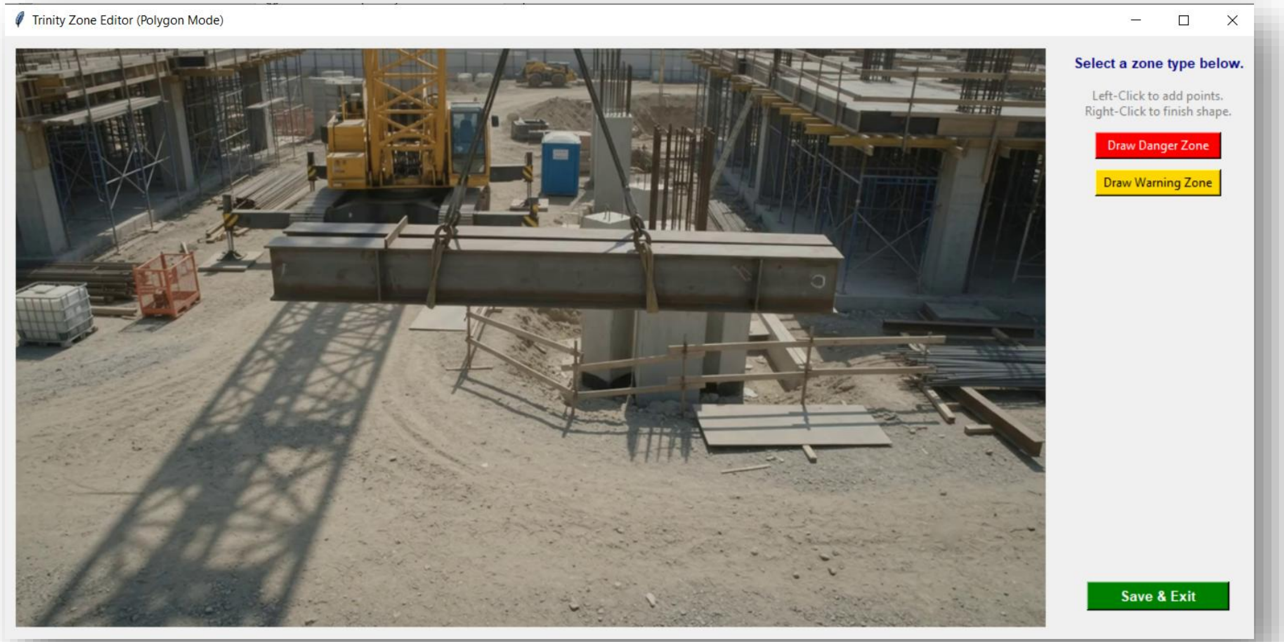
- **Setup (Human-in-the-loop):** Safety Officer manually draws zones via Web UI → Saves to zones.json.
- **Clean Input:** System reads *Raw Frames* to prevent visual noise.
- **Inference:** YOLO26n model detects all objects (Workers & PPE).
- **Smart Parsing:** Detections are split into 2 branches:
  - **Person Branch:** Tracks worker's foot coordinates.
  - **Standalone PPE Branch:** Fallback for disconnected items

A raw violated frame

5.2. Workflow

PHASE 1: INPUT & AI DETECTION





Zone drawing UI

5.2. Workflow

PHASE 1: INPUT & AI DETECTION

*Drawn zones (Red: **Danger**, Yellow: **Warning**)*

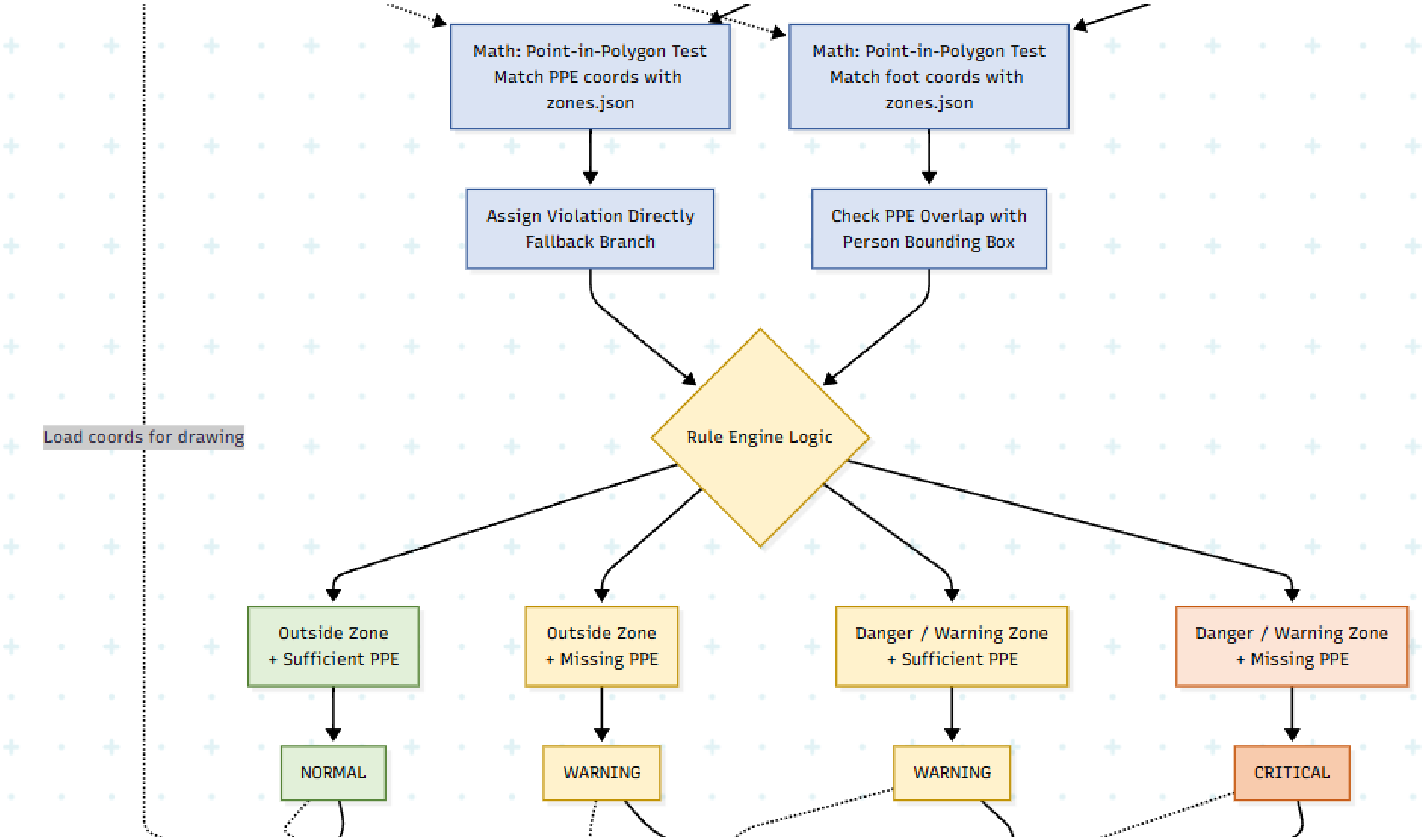
5.2. Workflow

PHASE 2: SPATIAL MATH & RULE ENGINE

- **Spatial Math (Point-in-Polygon):** Checks if coordinates (Person's foot OR Standalone PPE) fall inside zones.
- **Dual-Branch Logic:**
  - **Person Branch:** Analyzes overlap between Person and PPE bounding boxes.
  - **Standalone PPE Branch:** Directly assigns violation for isolated PPEs in restricted zones.
- **Rule Engine:** Classifies risk into 3 levels (CRITICAL, WARNING, NORMAL).

5.2. Workflow

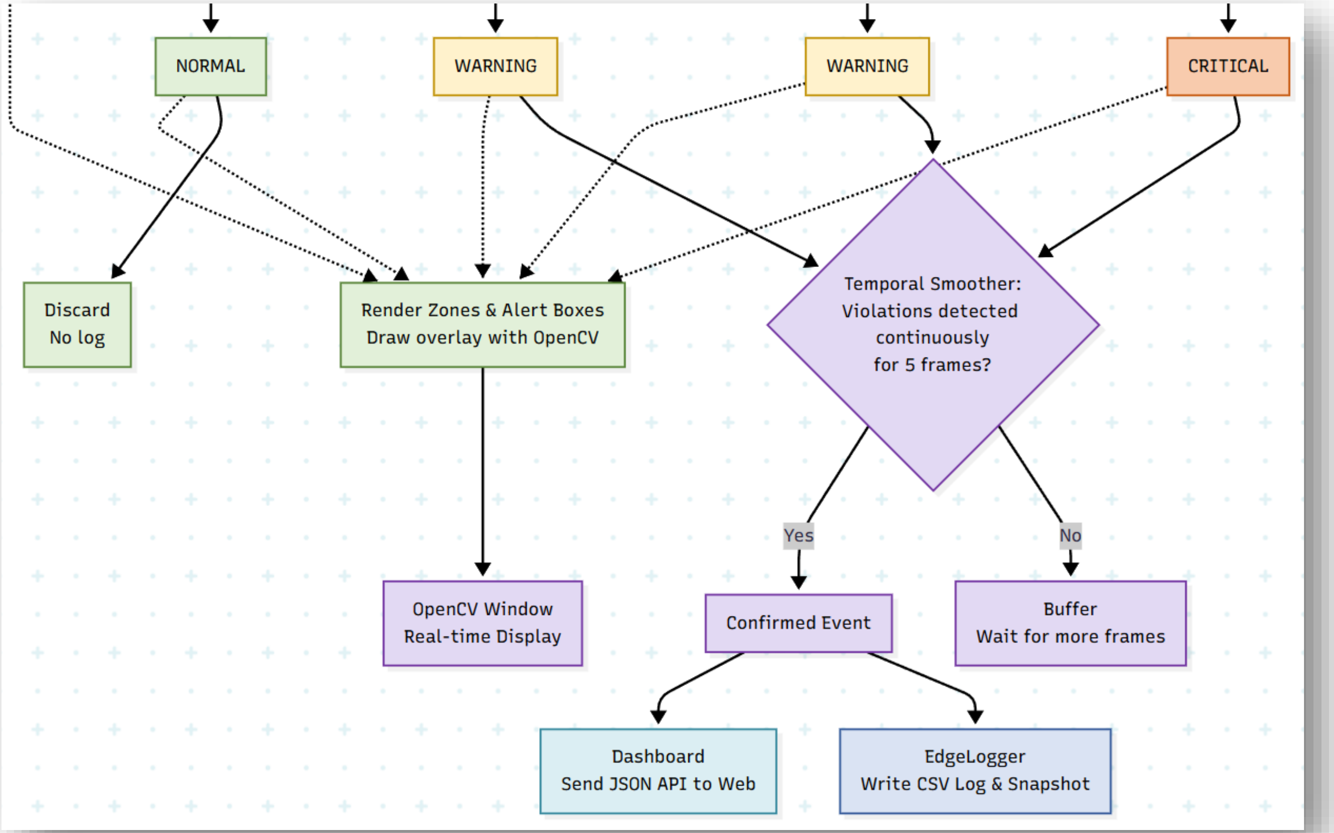
PHASE 2: SPATIAL MATH & RULE ENGINE



5.2. Workflow

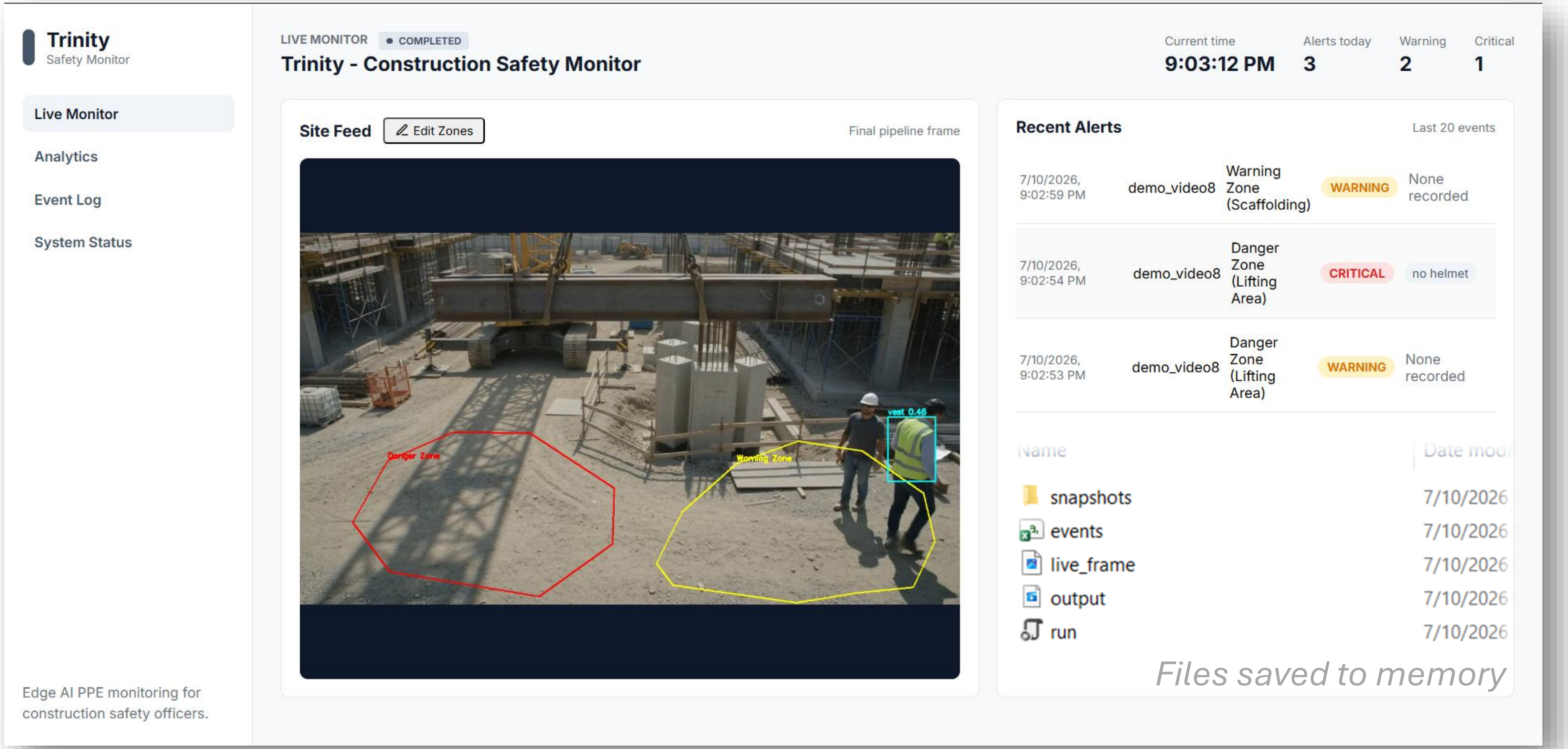
PHASE 3: EVENT FILTERING & OUTPUT

- **Temporal Smoother:** Anti-spam filter. Requires 5 consecutive frames to validate an event.
- **Data Logging:** 'Confirmed Events' are saved to local CSV and pushed to Web Dashboard via API.
- **Real-time Rendering:** OpenCV visually draws zones and alert boxes instantly (Bypasses the smoothing filter).



5.2. Workflow

PHASE 3: EVENT FILTERING & OUTPUT



Event outputs

5.2. Workflow

PHASE 3: EVENT FILTERING & OUTPUT

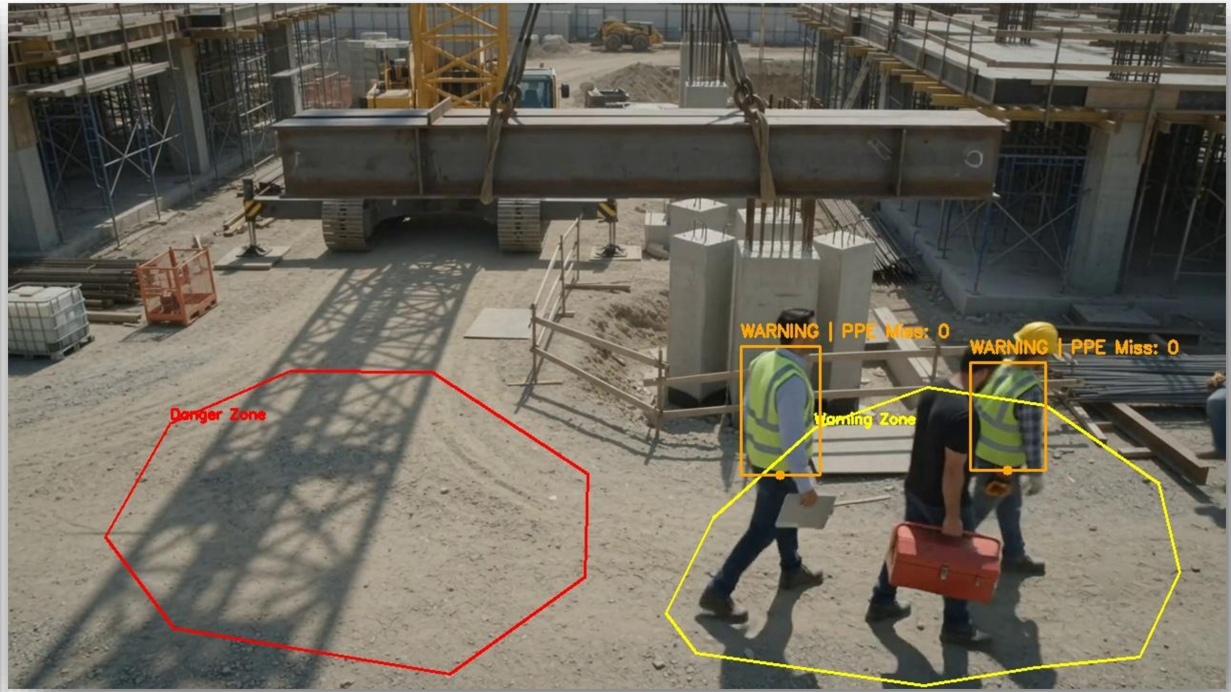
timestamp	camera_id	frame_id	zone_id	alert_level	violation_ty	confidence	bbox	snapshot_path
02:53.6	demo_vide	28	Z01	WARNING	None	0.64	[271, 362,	E:\3FIE\!Contest outside\Edge AI\Project\Trinity_Safety_System\edge-pipeline\logs\runs\20260710_210229_758914_demo_video8\snapshots\snap_camdemo_video8_f28_
02:54.7	demo_vide	52	Z01	CRITICAL	no_helmet	0.42	[376, 351,	E:\3FIE\!Contest outside\Edge AI\Project\Trinity_Safety_System\edge-pipeline\logs\runs\20260710_210229_758914_demo_video8\snapshots\snap_camdemo_video8_f52_
02:59.2	demo_vide	169	Z02	WARNING	None	0.72	[770, 360,	E:\3FIE\!Contest outside\Edge AI\Project\Trinity_Safety_System\edge-pipeline\logs\runs\20260710_210229_758914_demo_video8\snapshots\snap_camdemo_video8_f16

CSV Output

Violation detected

5.2. Workflow

PHASE 3: EVENT FILTERING & OUTPUT



CRITICAL *detected &* ***WARNING*** *detected*

6. RESULTS

6. Results

Model Format	Model Size	Latency (ms/img)	Throughput (FPS)	Overall mAP@0.5	Violation mAP@0.5
ONNX FP32 (Dynamic B=8)	10.3 MB	**7.61**	**131**	0.749	**0.671**
ONNX FP16 (Dynamic B=4)	**5.7 MB**	9.05	110	0.750	0.671

Table: ONNX Quantization Results

The **ONNX FP32** format with dynamic batching was selected as the **optimal deployment configuration**, achieving a high inference throughput of **131 FPS**. This format perfectly balances real-time edge processing speed with a reliable violation detection accuracy of **0.671 mAP@0.5.**

6. Results

Optimization Target	Baseline State	Edge-Optimized Result
Alert Reliability (baseline_yolo26n ONNX FP32)	124 alerts (No temporal filter)	6 alerts (N = 5 frames)
Network Efficiency	1.897 GB/hr (Raw video)	1.35 MB/hr (Logs + Snapshots)*

Table: Edge Optimization Result

- Implementing a **5-frame temporal smoothing** algorithm significantly improved alert reliability by reducing transient false alarms by **94.6%.**
- Transmitting **only event logs and snapshots** rather than raw video slashed bandwidth consumption by **99.93%**, enabling stable operation over standard mobile networks.

***Note:** Network efficiency assumes 1 event/min.*

6. Results

Best results summary

01

mAP@0.5

Violated classes:

no_helmet **0.871**

no_goggle **0.877**

no_gloves **0.810**

(yolo26n_base_onnx_f32 B=8)

02

Edge-Optimized

> Saved **99.93%** Bandwidth

> Reduced **94.6%** False positives

> Achieved **131 FPS** Edge throughput

03

E-2-E Pipeline

Context-Aware Logic
Worker & PPE status

Live Dashboard
Real-time telemetry, snapshot logs, and zone mapping.

**E-2-E: End-to-end*

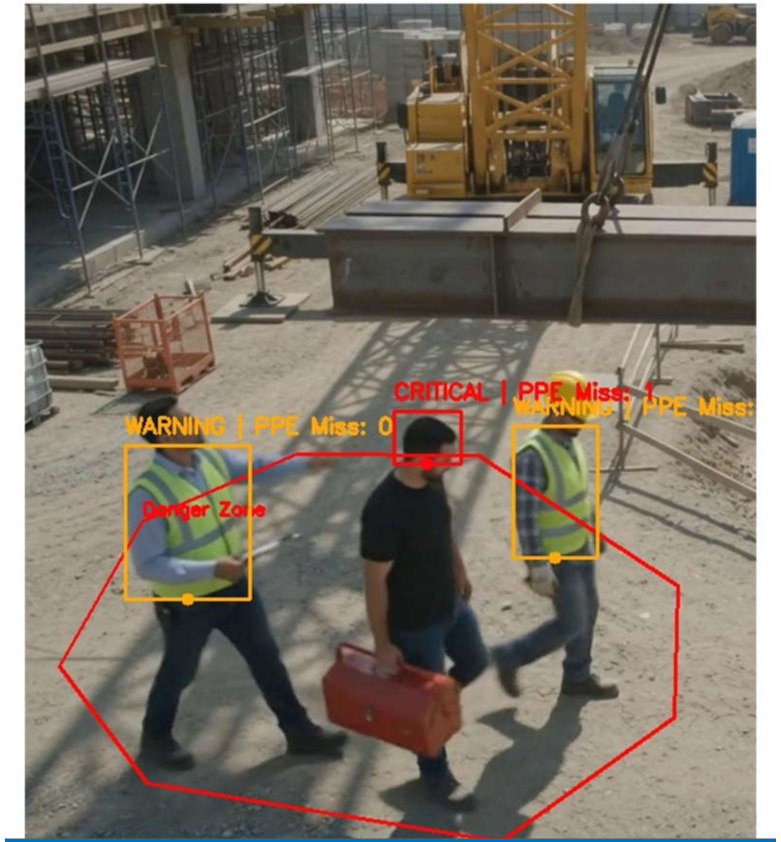
7. Conclusion

- Successfully deployed an Edge AI pipeline for real-time PPE monitoring, eliminating cloud-related latency, bandwidth, and privacy issues.

- The optimized baseline_yolo26n model (ONNX FP32) achieves 131 FPS at 2.93 ms/frame, enabling concurrent multi-camera processing.

- A 5-frame temporal smoothing algorithm reduces false alarms by 94.6%. The edge-first architecture reduces bandwidth consumption by 99.93% (1.35 MB/hour).

- ***Future Works:*** Enhance no_boots detection, integrate BIM for dynamic safety zones, and apply multi-modal sensor fusion for poor weather conditions.



CONSTRUCTION SITE SAFETY

8. APPLICATION DEMO

THANK YOU FOR LISTENING!

Q&A

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